Stop Calculating Correlations, Start Calculating Market Curvature

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LinkedIn post snapshot

Most quant systems trade noise and call it a strategy.
Not because the algorithm is bad.

Because the model is flawed from the start.
The standard approach: take prices, build a correlation matrix, and throw it into ML.

Sounds reasonable. It works poorly.
Why?
Because you're throwing away structure—the only thing that really matters.
The instruments aren't independent. They're linked by cross-impact, margin, and routing constraints.
These aren't scalar variables—they're sections of a vector bundle over a curved manifold.
Correlation is a flat approximation of a geometric curve.

Curvature is what you're actually trading.
We stopped estimating correlation and started calculating the Yang-Mills curvature directly. Order book → dynamic graph → Riemannian manifold in the continuum limit.
Order flow → Hodge decomposition → three orthogonal components.
Only one of them is traded. Co-exact: divergence-free loops, topologically protected from diffusion.
This is an arbitrage signal. Not an indicator. Not a pattern. A book topology invariant.
A full cycle—a snapshot of the order book before the signal—takes less than 1 ms on a single GPU.

No ML. No lookahead. Pure linear algebra on the device.
When a trader sees a "correlation breakdown," they see noise.
At this point, the algorithm already calculates non-zero holonomy and opens a position.
Open research. Link in profile.
Stop calculating correlations. Start calculating market curvature.

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